Project - Traditional Data Analysis

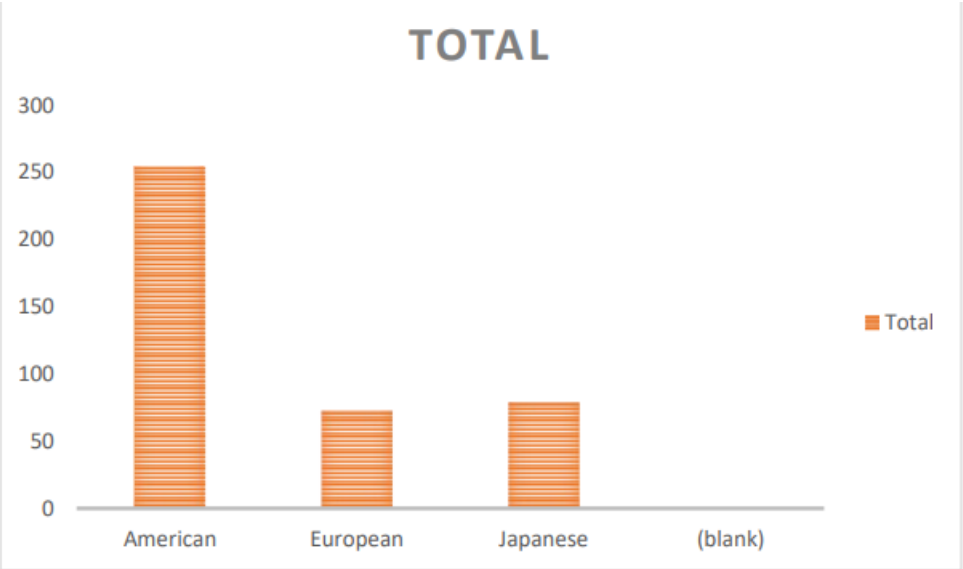
In these projects I have used cereals and cars datasets.

<u>1. Insights in the car's dataset: -</u>

Insight 1: -

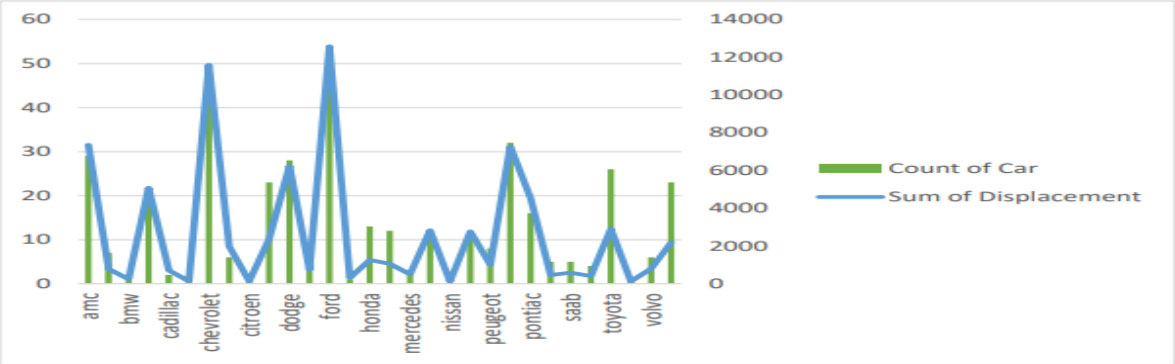
- ✓ A total of the mentioned car manufacturers operate in each country exists as shown above.
- ✓ The goals were reached by using pivot tables in combination with count functions in value field settings. Row Labels Count of Manufacturer American 254 European 73 Japanese 79 (blank) Grand Total 406.

Row Labels	Count of Manufacturer
American	254
European	73
Japanese	79
(blank)	
Grand Total	**406**



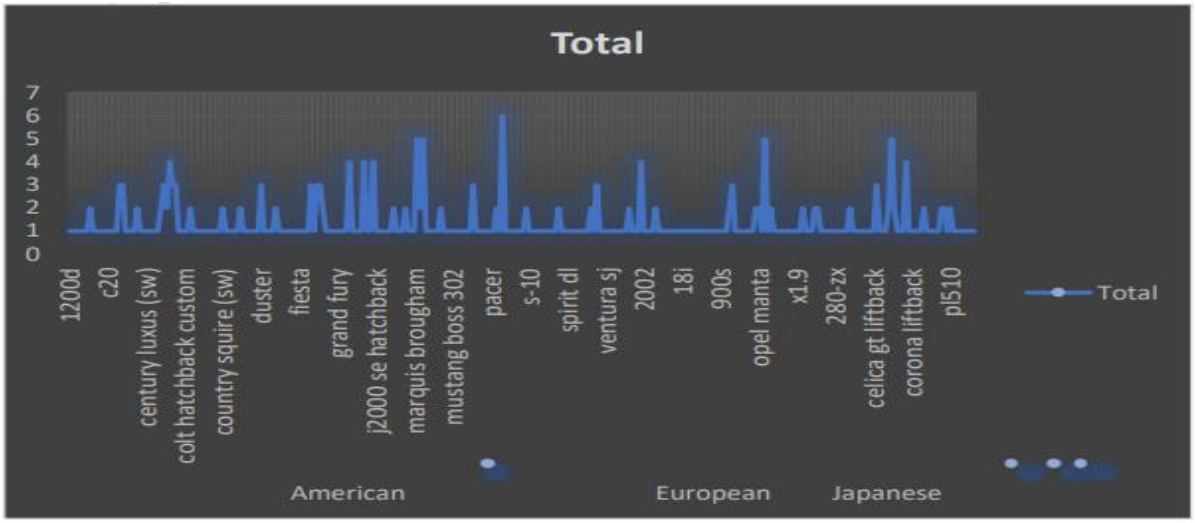
Insight 2: -

- ✓ Each company presented its total car production using this specific approach
- ✓ An analysis using pivot tables enabled the counting of cars in the column with the help of the count function.



Insights 3: -

- ✓ The total car figures per company within each area formed the typical number of vehicles.
- ✓ The average calculations for column displacement were possible through pivot tables.



Insights 4: -

- ✓ The initial process involved sorting the manufacturer column under AMC before sorting the horsepower data from high to low values.
- ✓ First, I arranged the manufacturer column to display AMC and then arranged the horsepower column from largest to smallest while creating the graph.

Insights 5: -

- ✓ The analysis included both the total number of produced cars and the manufacturer values from the provided dataset.
- ✓ The Manufacturer column served as a basis to create another column that utilized count function. ➢ We obtain the count using function count to accomplish subsequent large-to-small sorting followed by generation.

Difficulties: -

- MPG contains NA values that require replacement between zero and another value for the current dataset.
- Manually checking the values becomes hard because the data contains either complete numerical or no values at all.
- Different types of data require difficult scaling procedures
- The process of producing multiple column graphs using this data set proves to be an arduous task particularly while employing the current methods.
- The process of data cleaning proves to be very challenging.
- The processing of unorganized data in addition to broad datasets becomes highly complicated when working with Excel.

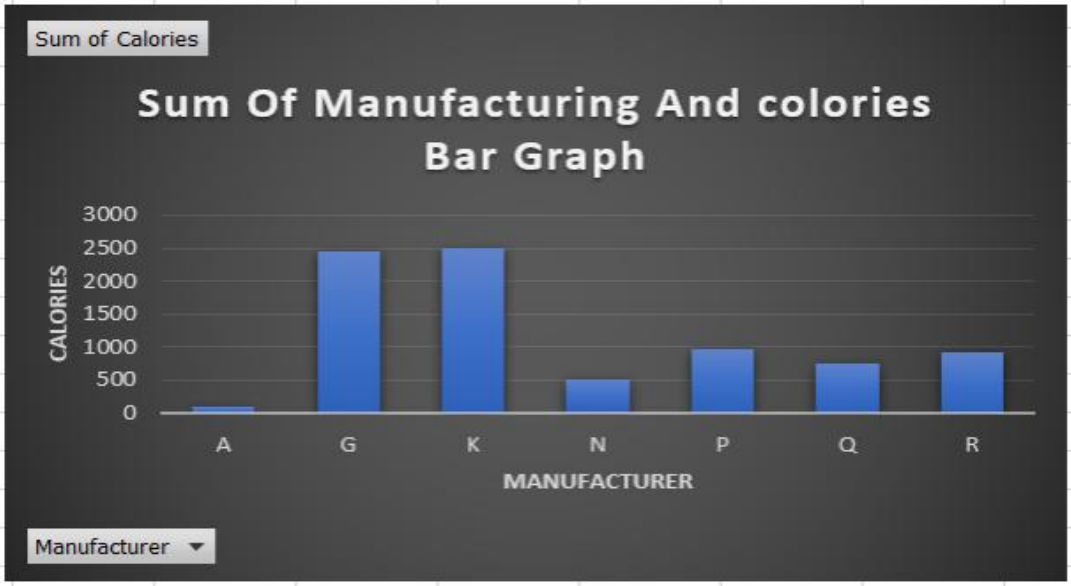
2. Insights in the cereal dataset: -

Insights 1: -

- ✓ Manufacturer "K" and "G" Produce the Most Calories
- ✓ Method Used: Pivot Table
- ✓ Created a Pivot Table with:

- ✓ Manufacturer in Rows
- ✓ Calories (Sum) in Values
- ✓ Total calories reach their peak in products manufactured by
- ✓ Companies K and G.

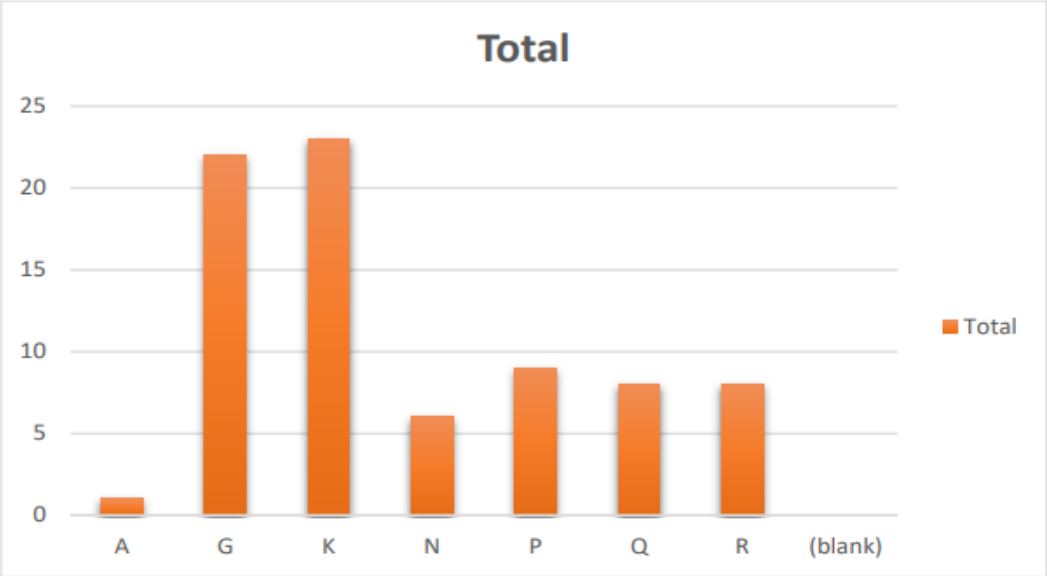
Row Labels	Sum of Calories
A	100
G	2450
K	2500
N	520
P	980
Q	760
R	920
Grand Total	8230



Insights 2: -

- ✓ Every Manufacturer generates different numbers of cereals.
- ✓ The pivot table function enables us to select the Manufacturer name alongside developing an Applied Count of cereals.

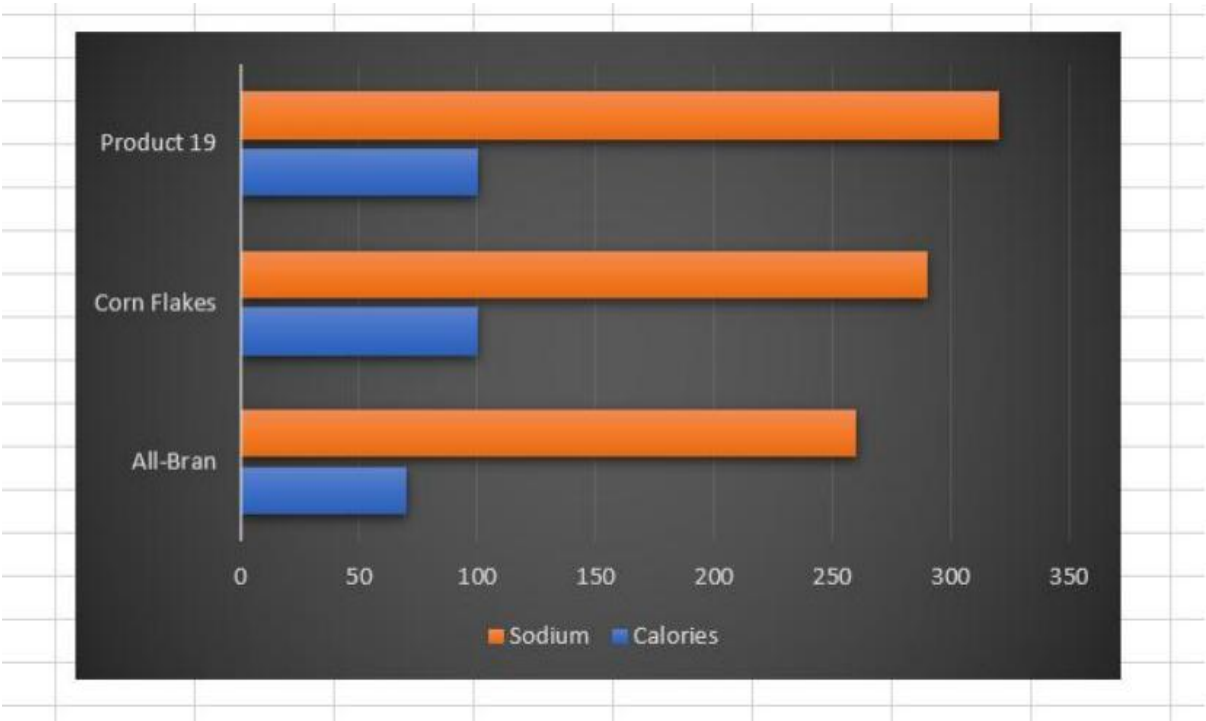
Row Labels	Count of Cereal
A	1
G	22
K	23
N	6
P	9
Q	8
R	8
(blank)	
Grand Total	**77**



Insights 3: -

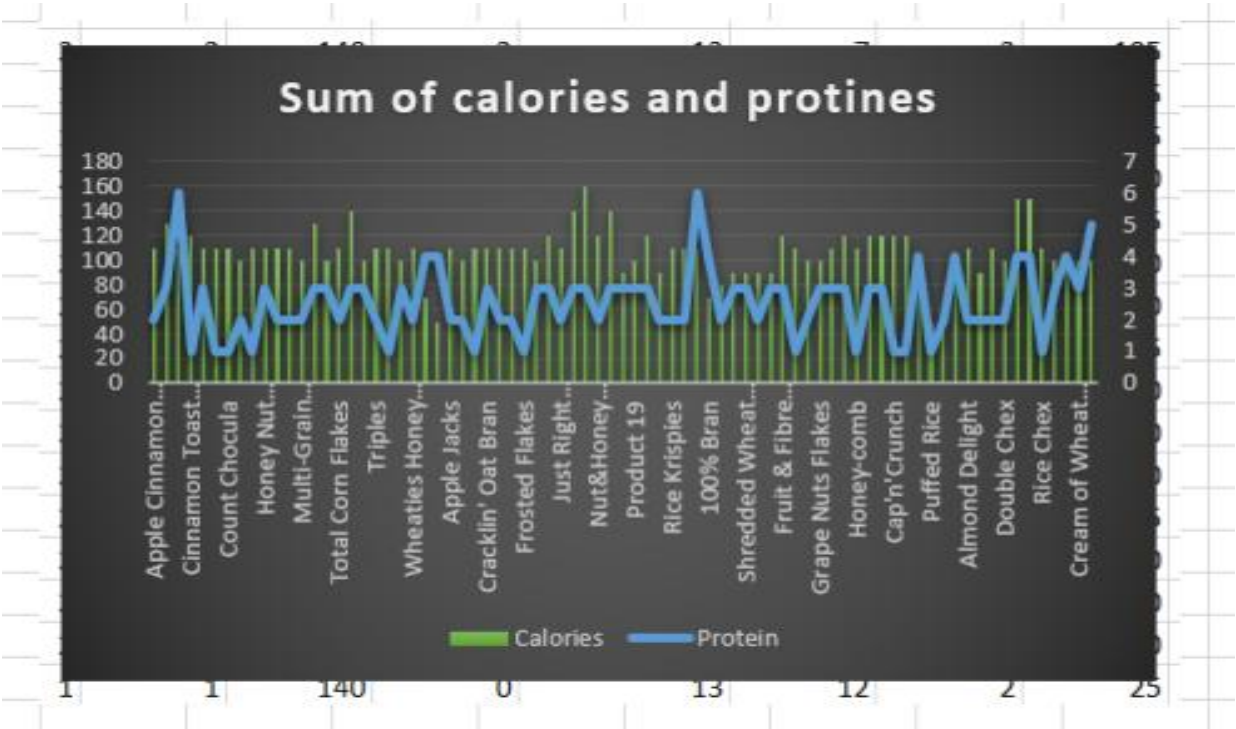
- ✓ The cereals All-Barn, Corn Flakes, Product 19 contain the highest sodium content among all products tested.
- ✓ By sorting sodium levels from largest to smallest I gained this information.

Cereal	Calories	Sodium
All-Bran	70	260
Corn Flakes	100	290
Product 19	100	320



Insights 4: -

- ✓ Calorie to Protein Ratio of each cereal in the data
- ✓ The pivot table and sorting technique allowed me to reach this goal.



Insights 5: -

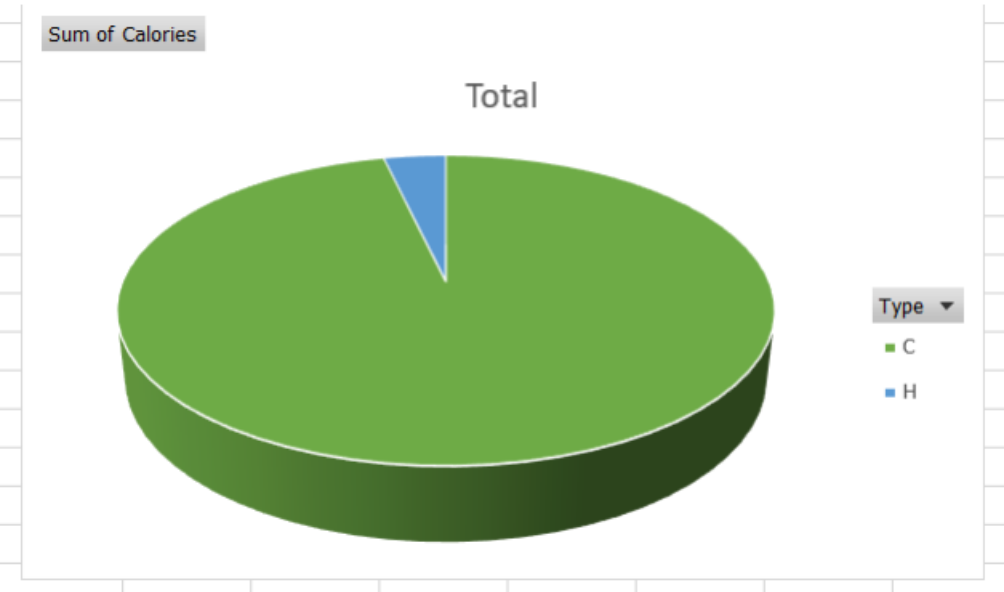
- ✓ The market contains two categories of Cereals which are Hot and Cold.

• Hot

• Cold

- ✓ The consumption of cold type cereals leads people to consume greater calories.
- ✓ My discovery of this insight was possible through using sum function in the pivot table within excel.

Row Labels	Sum of Calories
C	7930
H	300
Grand Total	8230



Difficulties: -

- The analysis of data in excel resulted in several challenges that I encountered
- I need to learn techniques to display the extensive cereal database through their carbohydrate and protein content.
- clearly without any clutter.
- There are only few restricted visualization methods available for displaying data to manufacturers and cereals.
- The large number of entries malfunctioned during excel file loading.
- I faced intense difficulty when cleaning the data through using this data set within an excel environment.
- It becomes difficult to adjust data values when calories along with proteins and other elements exist in varied numerical ranges.
- The alterations made on excel produce unsatisfactory visual outcomes during this scenario.

Thank you,
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